Find the Error

Grade 7 · 7.RP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Mia rides 1/2 mile in 1/4 hour. What is her unit rate in miles per hour? Student's answer: 1/8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.RP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

3/4 pound of cheese costs $6. What is the price per pound? Student's answer: $4.50

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Mario walks 1 1/2 miles in 1/3 hour. What is his speed in miles per hour? Student's answer: 1/2 mile per hour

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is this table proportional? x = 1, 2, 3, 4 and y = 3, 6, 9, 10. Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

4 items cost $12 in total. What is the constant of proportionality for cost per item? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The equation is y = 2.5x. What is y when x = 8? Student's answer: 10.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.RP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A restaurant bill is $40. What is a 15% tip? Student's answer: $15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An $80 item increases in price by 25%. What is the new price? Student's answer: $105

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A $25 book has an 8% sales tax. What is the total price? Student's answer: $2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add -3 + 8. Student's answer: -11

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Subtract 4 - (-6). Student's answer: -2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add -2.5 + (-4). Student's answer: 6.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.NS.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply (-4) × (-6). Student's answer: -24

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide (−12) ÷ 3. Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply (-2/3) × (3/4). Student's answer: 1/2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A team gains 3 points, then loses 5 points. What is the net change? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The temperature falls from −4°F to −12°F. What is the change in temperature? Student's answer: 8°F

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A fish sinks 2.5 feet every minute for 4 minutes. What is its position change in feet? Student's answer: 10 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $3(x + 2) - x$. Student writes: $2x + 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify 2x + 5 - 4x + 1. Student writes: -2x + 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.EE.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $4(2y - 1) + 3$. Student writes: $8y + 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.EE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 4x + 10 in factored form. Student writes: 4(x + 2.5)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An item is 15% off a price p. Which expression shows the sale price? Student writes: 0.15p

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor 9y - 6. Student writes: 3(3y + 2)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Estimate 4.8 × 6.2. Student's estimate: 60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Four notebooks cost $6.75 each, plus $3 for tax. What is the total? Student's answer: $24

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 2x + 3 = 11. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 4x + 5 = 25. Student's answer: 7.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $3(x + 1) = 18$. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 2x - 7 = 9. Student's answer: 16

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A map scale is 1 cm = 5 km. Two cities are 3.5 cm apart on the map. How far apart are they?
Student's answer: 3.5 km

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A drawing has a scale of 1 inch = 4 feet. A room is 12 feet long. How long is it in the drawing?
Student's answer: 48 inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A scale drawing uses 1 cm = 2 m. A wall is 6 m long. How long is it in the drawing? Student's answer: 12 cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Can a triangle have sides of 3 cm, 4 cm, and 5 cm? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Can a triangle have sides of 2 cm, 3 cm, and 6 cm? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Can a triangle have angles of 60°, 60°, and 60°? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plane cuts a cube parallel to one of its faces. What is the shape of the cross-section?
Student's answer: triangle

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plane cuts a cylinder parallel to its bases. What is the shape of the cross-section? Student's answer: rectangle

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plane cuts through two opposite edges of a cube. What is the shape of the cross-section?
Student's answer: triangle

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the circumference of a circle with radius 3 cm. Student's answer: 9π cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a circle with diameter 10 cm. Student's answer: 100π cm²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a circle with radius 5 cm. Student's answer: 10π cm²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are supplementary. One is 65°. What is the other? Student's answer: 25°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are complementary. One is 35°. What is the other? Student's answer: 145°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are vertical angles. One is 70°. What is the other? Student's answer: 110°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has a base of 8 cm and a height of 5 cm. What is its area? Student's answer: 40 cm²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A circular table has a radius of 3 feet. What is its area, to the nearest hundredth? Student's answer: 18.85 ft²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A prism has a base area of 12 square units and a height of 5 units. What is its volume? Student's answer: 17 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coach asks 20 players on the basketball team their favorite sport to learn about the whole school. Is this sample representative? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 50 students is taken from a school of 500. Is the sample likely to be representative? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A survey asks the first 15 people in line at an ice cream shop about flavors for all the students. Is this sample representative? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a random sample of 50 students from a school of 800, 20 students said they like the new lunch menu. Estimate how many students in the school like it. Student's answer: 20

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a random sample of 100 students, 60 said yes to a question. Estimate the number of yes answers in a population of 1,500. Student's answer: 60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 25 students has 8 who like soccer. What is the estimated percent of the population who like soccer? Student's answer: 8%

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.SP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has data 10, 12, 14. Group B has data 11, 13, 15. Which group has the higher mean?
Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a range of 3 and Group B has a range of 10. Which group is more variable?
Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a median of 22 and Group B has a median of 33. What is the difference between the medians? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.SP.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group X has a mean of 12 and a mean absolute deviation of 2. Group Y has a mean of 15 and a mean absolute deviation of 2. Is the difference in means large compared to the variability?
Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 30 students from School A has a mean of 7 hours of sleep. A random sample from School B has a mean of 7 hours. Student says: The schools have different sleep amounts.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two random samples have medians of 40 and 42, with spread of 15 in each. Student says: The difference is meaningful.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A weather app says the chance of rain is 1.5. Can this be correct? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the probability of an impossible event? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.SP.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the probability of rolling a number less than 7 on a standard die? Student's answer: 0

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coin is flipped 50 times and lands heads 28 times. What is the experimental probability of heads? Student's answer: 0.28

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A spinner is spun 100 times and lands on red 20 times. What is the experimental probability of red? Student's answer: 20

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coin has a theoretical probability of heads of 0.5. A student flips it 10 times and gets 3 heads. Student says: The coin must be broken.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.SP.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A fair six-sided die is rolled. What is the probability of rolling a 4? Student's answer: 1/3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model gives outcomes A, B, C, and D with probabilities 0.2, 0.3, and 0.4 for A, B, and C. What is P(D)? Student's answer: 0.2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A spinner has 8 equal sections, and 3 are blue. What is P(blue)? Student's answer: 3/5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two fair coins are flipped. What is the probability of getting two heads? Student's answer: 1/2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.SP.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two fair dice are rolled. What is the probability that the sum is 7? Student's answer: 1/12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 7 · 7.SP.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coin is flipped and a die is rolled. How many total outcomes are there? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.